Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/03/2016
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	Impact of Space Radiation Induced Neuroinflammation on Alzheimer and Parkinson Disease Pathology		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	11/13/2012	**End Date:**	10/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2016 per S. Monk/LaRC (Ed., 12/11/15)		
Key Personnel Changes/Previous PI:	September 2014: Lisa Opanashuk recently took a position with the FDA and is no longer a University of Rochester Faculty member. We are identifying a new collaborator to assist with analysis of the PD mouse brain sections. September 2013: Sean Hurley was listed in this record as a Co-Investigator; however, Sean left the University prior to the start of this grant and was probably inadvertently included because of the delay between the initial application and the actual funding date. I removed his name from the Co-Investigator section.		
COI Name (Institution):	Olschowka, John (University of Rochester Medical Center) Williams, Jacqueline (University of Rochester Medical Center)		
Grant/Contract No.:	NNX13AC33G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The brain reacts to injury by activation of glial cells and expression of inflammatory mediators in a process called neuroinflammation. Neuroinflammation occurs in the setting of age-associated neurodegenerative diseases such as Alzheimer's (AD) and Parkinson's (PD) and numerous studies suggest a contributory role to disease pathogenesis. We have previously shown that space radiation, at doses approaching those that will be experienced by astronauts on long missions to the moon and Mars, elicits brain neuroinflammation. Therefore we propose to investigate the possibility that space radiation influences pathological changes associated with neurodegenerative diseases. Our studies will utilize three different mouse models of the two most prevalent age-associated neurodegenerative diseases (Alzheimer's and Parkinson's) to assess the effects of space radiation on disease-associated pathology and behavioral changes as well as indices of neuroinflammation and oxidative stress. Findings from these studies will help to refine risk estimates for future space explorers and may offer possible strategies for mitigation of such risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this third year of the grant we continued to make progress in both Aims 1 and 2. As mentioned previously, we arranged for production of sufficient transgenic mice representing both neurodegenerative models (AD and PD) with Jackson Laboratories and in October and November 2013 successfully carried out three separate irradiation campaigns during NSRL-13C to initiate Aim 1. The specific dates and beams were: 1) 300 MeV/µ Silicon at 10 and 100 cGy on October 31; 2) 600 MeV/µ Iron at 10 and 100 cGy on November 15; and 3) 200 cGy mixed protons mimicking a solar flare on November 14. For each ion and energy 10 male and 10 female mice were irradiated. Two set of sham irradiated (control) mice with 10 males and 10 females in each set were included in the experiment, for a total of 140 transgenic mice. Mice were approximately 7 months of age at the time of irradiation and were shipped back to Rochester to be aged. Notably, the mice were sacrificed at 16-17 months of age, which is a little younger than planned (original plan was 18 months of age), in order to have sufficient number of male and female mice for these studies. Indeed, we experienced some degree of unexpected death, particularly in males, which does not appear to be correlated to whether they were irradiated or not. Those mice were sacrificed near the end of the last reporting period and in this period we carried out a series of histological analyses, which are described below. In our earlier studies using a different AD mouse model (APPswe/PS1dE9) we found enhanced amyloid beta plaque accumulation and behavioral deficits following 100 cGy 56Fe particles (1 GeV/µ; Cherry et al., 2012. PLoS one 7:e53275). To determine if the triple transgenic mouse model might show similar exacerbation of pathology with irradiation, we carried out immunohistochemical staining of tissue sections with 6E10. Although Aß plaques were readily observed in female mice in this cohort, we did not detect Aß deposits in male mice in any brain region. In females we observed deposits of Aß in the subiculum, septal nuclei, and in the cortex. Regional analysis of plaque load revealed only one significant change: an increase in cortical Aß load in female mice exposed to 10 cGy iron (600 MeV/µ). We also examined the effects of radiation on tau phosphorylation in 3xTg mice using an antibody against p205, an epitope that is prominently phosphorylated in AD brain. Similar to the findings with Aß, p205 staining was increased only in the female mice exposed to 10 cGy iron. We also increased staining for ICAM-1 in brains of female mice exposed to 10 cGy iron. Taken together, these findings in the 3xTg mouse model of AD illustrate differences between sexes with regard to presence of pathology and a possible increase in Aß pathology and associated ICAM-1 and tau phosphorylation with 10 cGy, but not 100 cGy iron exposure. Notably, our immunohistochemical results show significantly variability within mouse groups, which increases the challenge of finding significant differences between different exposure groups. Nevertheless, it seems clear that the various ions used (proton, silicon, and iron) gave distinct responses. With regard to Aim 2, we contracted with Jackson to do a rapid derivation of Parkin-Q311X mice with a goal of 80 mice (40 male, 40 female) harboring the disease model transgene and 80 control littermates. These mice were successfully bred, aged at Jackson Laboratories, and shipped to Brookhaven National Laboratory (BNL) to be utilized during the Spring 2014 NASA Space Radiation Laboratory (NSRL) run (NSRL14A). The specific dates and beams were: 1) 300 MeV/µ Silicon at 50 cGy on May 1; 2) 600 MeV/µ Iron at 50 cGy on May 12; and 3) 100 cGy mixed protons mimicking a solar flare on May 13. These animals were approximately 8 months of age at the time of irradiation and were shipped to Rochester. Prior to sacrifice at 18 months of age, mice underwent behavioral tests that are sensitive to basal ganglia dysfunction. The specific tests included activity recording, rota-rod, and tape removal tests. The behavioral data sets gathered were extensive and are still being analyzed. During this reporting period we carried out blinded densitometric analysis of striatal TH staining, which showed evidence for reduced fiber staining in male, but not female mutant transgenic mice that was not influenced by radiation. Additional histological analyses are underway.
Bibliography Type:	Description: (Last Updated: 01/07/2026)
Abstracts for Journals and Proceedings	O'Banion MK, Olschowka JA, Williams JP. "Space radiation induced neuroinflammation on Alzheimer and Parkinson disease pathology." Invited speaker, 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	O'Banion MK. "HZE Dose Effects on Neuroinflammation and Neurogenesis: An Evaluation of Central Nervous System Responses to Space Radiation." Invited Speaker, Space Radiation and Heavy Ions in Therapy Symposium 2015, 7th International Workshop on Space Radiation Research, Osaka, Japan, May 23-24, 2015. Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015. , May-2015
Abstracts for Journals and Proceedings	O'Banion MK. "HZE Exacerbates Alzheimer's Disease Pathology in Transgenic Mouse Models." Invited Speaker, 15th International Conference of Radiation Research (ICRR), Kyoto, Japan May 25-29, 2015. 15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015. , May-2015